Data Technician Workbook Week 3 Databases and SQL

Name:

Course Date:

Table of contents

Day 1: Task 1 3

Breakout Room Activity: Designing a Relational Database (45 min) 3

Business Scenario 3

Day 1: Task 2 6

Day 2: Task 1 8

Day 3: Task 1 11

Day 3: Task 2 13

Day 4: Task 1: SQL Practical 15

Additional Resources 19

Additional Information 19

Course Notes 20

Day 1: Task 1

Breakout Room Activity: Designing a Relational Database (45 min)

Business Scenario

You have been hired as a junior data technician for a **small training company** called **SkillUp Academy**. The company currently stores all its data in a single spreadsheet, which is causing duplicate information, data inconsistencies, and severe difficulty updating records.

Your team's task is to collaboratively design a simple relational database that organizes this data more effectively.

⏱️ Part 1: The Team Huddle (10 Mins)

Task 1: Database Purpose Discuss the scenario as a team. Work together to write a brief, 2-to-3-sentence "Elevator Pitch" explaining the purpose of the new database.

✍️ Your team response must include: What the database will be used for, who will use it, and what type of data it will store.

Your Team's Purpose Statement:

The SkillUp Academy database will be used to store and manage information in one organised and reliable system, reducing duplicate records, avoid inconsistencies and making updates easier. It will be used by admins, students, trainers and management to manage data such as learners, courses, trainers, enrolments and attendance.

Part 2: Identify Tables and Columns (15 Mins)

Task 2: Database Schema Blueprint Brainstorm the four main tables SkillUp Academy needs (e.g., the core entities they track). For each table, agree on a unique Primary Key and at least 3-4 essential columns.

Fill out the team blueprint below:

📋 Table 1: Learner

- Primary Key: LearnerID
- Columns:
  - First Name
  - Last Name
  - Email
  - Phone Number
  - CourseID (Foreign Key)

📋 Table 2: Teacher

- Primary Key: TeacherID
- Columns / Attributes:
  - First Name
  - Last Name
  - Email
  - Course they teach

📋 Table 3: Course

- Primary Key: Course ID
- Columns / Attributes:
  - Course Name
  - Course Level
  - Topics
  - Length
  - Price
  - TeacherID

📋 Table 4: Enrolment

- Primary Key: EnrolementID
- Columns:
    - LearnerID
    - CourseID
    - TeacherID
    - Enrolement Date
    - Completion status

Part 3: Define Relationships (10 Mins)

Task 3: Connecting the Data Discuss how your tables talk to each other. Identify at least two one-to-many relationships required to make this database function correctly.

1. Relationship 1:
    - Table A: __Students__ connects to Table B: __Course__
    - Which column is the Foreign Key? _CourseID_
    - Why is this relationship needed? This Relationship is needed to show what students are on each course and to know what course each student is on.
2. Relationship 2:
    - Table A: __Teachers__ connects to Table B: __Course__
    - Which column is the Foreign Key? _TeacherID_
    - Why is this relationship needed? One teacher can teach multiple courses, so this allows skillup to track which trainer is responsible for each course without repeating the trainer's personal details.

Day 1: Task 2

Choosing the Right Type of Database
Business Scenario

SkillUp Academy is planning to expand its systems.
In addition to structured course and learner data, the company now wants to store:

- Website click data
- User activity logs
- Feedback comments from learners
- Uploaded documents

They are unsure whether a **relational database** or a **non-relational database** would be more suitable for this data.
You have been asked to help explain the options.

Task 1: Compare Database Types
In your own words, explain the difference between:

- A **relational database**
- A **non-relational (NoSQL) database**

Your answer should include:

- How data is stored
- How structured the data is
- An example use case for each

✍️ *Write in clear, simple language.*

Relational database :

A relational database stores data in **structured tables made up of rows and columns. The tables can be connected using primary and foreign keys, making it easy to organise and manage consistent data. It is suitable for data with a fixed structure, s**uch as learner records, courses and enrolments.

Example use case: A training company could use a relational database to store learner details, course information, trainers and enrolments.

Non-relational (NoSQL) database :

A non-relational database does not rely on fixed tables and can store data in formats such as documents, key-value pairs or collections. It is more flexible and can handle unstructured or changing data, such as website activity, feedback comments, logs and uploaded documents.

Example use case: SkillUp Academy could use a NoSQL database to store website click data, user activity logs and learner feedback, where the information may vary in structure.

Task 2: Identify Suitable Data for a Non-Relational Database

From the scenario above, identify **two types of data** that would be better stored in a **non-relational database**.

Examples may include:

- User activity logs
- Feedback text
- Website events

For each type of data:

- State **what the data is**
- Explain **why a non-relational database is suitable**

✍️ *Focus on flexibility, structure, and data format.*

1. User Activity Logs :

- **What the data is:** Records of actions users take on the SkillUp Academy website or system, such as logins, page views, searches and course activity.
- **Why a non-relational database is suitable:** Activity logs can contain different types of information and can change over time. A NoSQL database can store this flexible and varied data without requiring every record to have the same structure.

2. Feedback Comments :

- **What the data is:** Comments and written feedback submitted by learners about their courses or learning experience.
- **Why a non-relational database is suitable:** Feedback can be unstructured and vary greatly in length and content. A NoSQL database can store text and additional information alongside it without needing a fixed set of columns for every piece of feedback.

Task 3: Justify Your Choice

Explain **why this data would not be ideal** for a traditional relational database.

Consider:

- Data structure
- Frequency of changes
- Volume of data

✍️ *Use plain English — no advanced terminology required.*

User activity logs and feedback comments would not be ideal for a traditional relational database because the data can vary in structure and may change or grow frequently. A large website can generate a high volume of activity logs, while feedback can be different lengths and formats. A non-relational database provides more flexibility for storing this type of changing and varied data without requiring a fixed structure for every record.

Day 2: Task 1

You are a junior data analyst at **Northwind Traders**. Your manager has assigned you a series of real-world business tasks. For each scenario:

1. **Write an SQL query** to solve the request. Write an SQL query to solve the request.
2. Run the query in Supabase using the Northwind database.
3. Paste your query in the box below each question as evidence of completion.

---

1. 📘 *Retrieve Full Customer Data*

Your manager has asked you to export the full list of all customer details into a report.

➡️ **Write a SQL query to retrieve all columns from the Customers table.**

```
SELECT * FROM Customers;
```

---

2. 📘 *Customer Names and Cities for Marketing*

A marketing analyst is targeting a campaign based on customer names and their cities.

➡️ **Write a SQL query to retrieve only the CustomerName and City columns from the Customers table.**

```
SELECT CustomerName, City FROM Customers;
```

---

3. 📘 *Unique Cities for Delivery Network Expansion*

The logistics team wants to know all the different cities where customers are located (no duplicates).

➡️ **Write a SQL query to retrieve distinct values from the City column in the Customers table.**

```
SELECT DISTINCT City FROM Customers;
```

---

4. 📘 *High-Value Products Report*

The product manager wants to analyse products priced over £50.

➡️ **Write a SQL query to retrieve all columns from the Products table where the Price is greater than 50.**

```
SELECT * FROM Products WHERE Price > 50;
```

---

5. 📘 *International Customers Targeting (USA & UK)*

The marketing team is preparing campaigns for customers in the USA and UK.

➡️ **Write a SQL query to retrieve all columns from the Customers table where the Country is either 'USA' or 'UK'.**

```
SELECT * FROM Customers WHERE Country IN ('USA', 'UK');
```

---

6. 📘 *Recent Orders Report*

Your team needs to analyze the latest orders.

➡️ **Write a SQL query to retrieve all columns from the Orders table, sorted by OrderDate in descending order.**

```
SELECT * FROM Orders ORDER BY OrderDate DESC;
```

---

7. 📘 *Mid-Range Products Listing*

You are preparing a product list for items priced between £20 and £50.

➡️ **Write a SQL query to retrieve all columns from the Products table where the Price is between 20 and 50, ordered by descending Price.**

```
SELECT * FROM Products WHERE Price BETWEEN 20 AND 50 ORDER BY Price DESC;
```

---

8. 📘 *Local Marketing in the US (Portland & Kirkland)*

You're focusing on customers from Portland and Kirkland in the USA.

➡️ **Write a SQL query to retrieve all columns from the Customers table where the Country is 'USA' and City is either 'Portland' or 'Kirkland', ordered by ascending CustomerName.**

```
SELECT * FROM Customers WHERE Country = 'USA' AND City IN ('Portland', 'Kirkland') ORDER BY CustomerName ASC;
```

---

9. 📘 *Customers from the UK or London*

Prepare a list for promotional emails targeting customers from the UK or those based in London.

➡️ **Write a SQL query to retrieve all columns from the Customers table where the Country is 'UK' or City is 'London', ordered by descending CustomerName.**

```
SELECT * FROM Customers WHERE Country = 'UK' OR City = 'London' ORDER BY CustomerName DESC;
```

---

10. 📘 *Product Inventory for Selected Categories*

The inventory team needs products from Category 1 or 2, sorted alphabetically.

➡️ **Write a SQL query to retrieve all columns from the Products table where the CategoryID is 1 or 2, ordered by ascending ProductName.**

```
SELECT * FROM Products WHERE CategoryID IN (1, 2) ORDER BY ProductName ASC;
```

Day 3: Task 1

Scenario-Based Activity: Understanding SQL JOINs

Business Scenario

SkillUp Academy stores its data across several related tables, such as:

- Learners
- Courses
- Trainers
- Enrolments

To create reports (for example, seeing which learners are enrolled on which courses), the data team needs to **combine data from multiple tables**. Different types of **SQL JOINs** are used depending on what information is needed.

Task 1: Explain JOIN Types (Plain English)

For each JOIN type below:

- Explain **what the JOIN does** in simple terms
- Explain **when it would be used** in a real scenario

You do **not** need to write SQL code unless instructed.

JOIN Types to Explain

- Inner Join
- Left Join
- Right Join
- Full Join
- Cross Join

✍️ *Use short, clear explanations.*

Join type:	What it does:	When to use:
Inner	Returns only the records that have a match in both tables.	To see which learners are enrolled on courses, showing only learners who have an enrolment.

Left	Returns all records from the left table, plus matching records from the right table.	To see all learners, including learners who are not currently enrolled on a course.
Right	Returns all records from the right table, plus matching records from the left table.	To see all courses, including courses that currently have no learners enrolled.
Full	Returns all records from both tables, whether or not they have a match.	To see all learners and all courses, including learners without courses and courses without learners.
Cross	Combines every record from one table with every record from another table.	To create every possible learner-course combination, for example when exploring potential course options.

Task 2: Match JOINs to Data Scenarios

Below are example reporting needs at SkillUp Academy.

For each one:

- Identify **which JOIN type** would be most suitable
- Explain **why**

Example Scenarios

1. Viewing only learners who are enrolled on a course
2. Listing all courses, including those with no enrolments yet
3. Showing all learners and all courses, even if they are not linked
4. Creating all possible combinations of learners and courses for planning
5. Comparing learners to other learners (e.g. same course, same city)
6. Viewing enrolments even if some related data is missing

✍️ *Explain your reasoning in plain English.*

#	scenario	Join type	Why?
1	Viewing only learners who are enrolled on a course	INNER	It only shows learners who have a matching enrolment, so learners without a course are left out.
2	Listing all courses, including those with no enrolments yet	LEFT	It keeps all courses in the results, even when there is no matching enrolment.
3	Showing all learners and all courses, even if they are not linked	FULL	It includes all records from both tables, whether or not they have a match.
4	Creating all possible combinations of learners and courses for planning	CROSS	It combines every learner with every course to create all possible combinations.
5	Comparing learners to other learners (e.g. same course, same city)	SELF	It joins a table to itself so records for different learners can be compared.
6	Viewing enrolments even if some related data is missing	LEFT	It keeps the enrolment records even when matching information in a related table is missing.

Day 3: Task 2

You are a junior data analyst at **Northwind Traders**. Your manager has assigned you a series of real-world business tasks. For each scenario:

1. **Write an SQL query** to solve the request. Write an SQL query to solve the request.
2. Run the query in Supabase using the Northwind database.
3. Paste your query in the box below each question as evidence of completion.

1. 📘 *Linking Products to Suppliers*

Management wants to know which supplier provides each product in the inventory.

➡️ **Write a SQL query to find the supplier of each product.**

```
SELECT Products.ProductName, Suppliers.CompanyName AS SupplierName
FROM Products
JOIN Suppliers ON Products.SupplierID = Suppliers.SupplierID;
```

---

2. 📘 *Classifying Products by Category*

The category manager is reviewing how products are organized.

➡️ **Write a SQL query to find the category of each product.**

```
SELECT Products.ProductName, Categories.CategoryName
FROM Products
 JOIN Categories ON Products.CategoryID = Categories.CategoryID;
```

---

3. 📘 *Category-Specific Product Report: Meat/Poultry*

The food department wants to view all items in the *Meat/Poultry* category.

➡️ **Write a SQL query to retrieve all products belonging to the Meat/Poultry category.**

```
SELECT Products.ProductName
 FROM Products
 JOIN Categories ON Products.CategoryID = Categories.CategoryID
WHERE Categories.CategoryName = 'Meat/Poultry';
```

---

4. 📘 *Complete Order Overview*

The business team wants to see a detailed order list with customer and employee information.

➡️ **Write a SQL query to retrieve the Order ID, Order Date, Customer Name, and Employee Name for all orders.**

```
SELECT Orders.OrderID, Orders.OrderDate, Customers.CompanyName AS CustomerName,
Employees.FirstName || ' ' || Employees.LastName AS EmployeeName
FROM Orders
JOIN Customers ON Orders.CustomerID = Customers.CustomerID JOIN Employees ON
Orders.EmployeeID = Employees.EmployeeID;
```

---

5. 📘 *Supply Chain Overview Report*

Your manager wants to see the product name, its category, and the name of its supplier all in one report.

➡️ **Write a SQL query to retrieve the Product Name, Category Name, and Supplier Name for all products.**

```
SELECT Products.ProductName, Categories.CategoryName, Suppliers.CompanyName AS
SupplierName
FROM Products
JOIN Categories ON Products.CategoryID = Categories.CategoryID JOIN Suppliers ON
Products.SupplierID = Suppliers.SupplierID;
```

---

6. 📘 *Yearly Order Summary – 1996*

The team is auditing customer orders made in 1996.

➡️ **Write a SQL query to create a report for all the orders of 1996 and their customers.**

```
SELECT Orders.OrderID, Orders.OrderDate, Customers.CompanyName AS CustomerName
FROM Orders
JOIN Customers ON Orders.CustomerID = Customers.CustomerID WHERE Orders.OrderDate >=
'1996-01-01' AND Orders.OrderDate < '1997-01-01';
```

---

7. 📘 *Product Count by Category*

The product team wants to know how many products exist under each category.

➡️ **Write a SQL query to retrieve all categories along with the number of products in each category.**

```
SELECT Categories.CategoryName, COUNT(Products.ProductID) AS ProductCount
FROM Categories
LEFT JOIN Products ON Categories.CategoryID = Products.CategoryID GROUP BY
Categories.CategoryName;
```

---

8. 📘 *Sales Volume Breakdown*

The sales department wants to analyze how much of each product was ordered and at what price.

➡️ **Write a SQL query to retrieve all products with their prices and the quantity ordered for each product.**

```
SELECT Products.ProductName, OrderDetails.UnitPrice, OrderDetails.Quantity
FROM Products
JOIN OrderDetails ON Products.ProductID = OrderDetails.ProductID;
```

Day 4: Task 1: SQL Practical

You are working as a **junior data analyst** for an international research organization analysing countries, cities, and languages from around the world. Your job is to write and run SQL queries that provide useful global insights.
For each task:

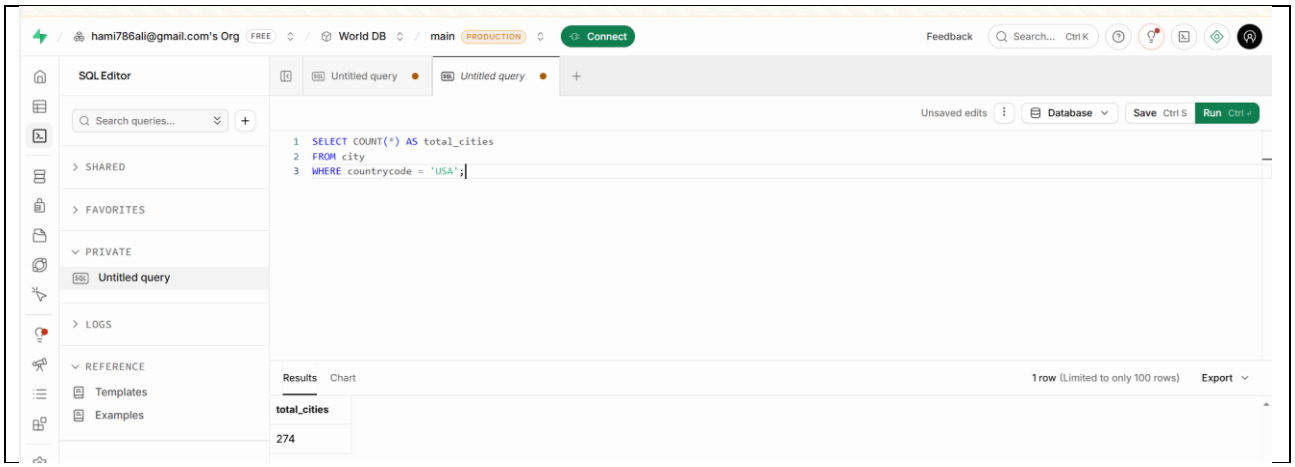
1. **Write the appropriate SQL query** to meet the scenario's objective.
2. **Run your query** using **MySQL Workbench** and the **world** database.
3. **Take a screenshot** that clearly shows both your SQL code and the result table.
4. **Paste the copy of your query in the box provided for each question as evidence of completion.**

Setting up the database:

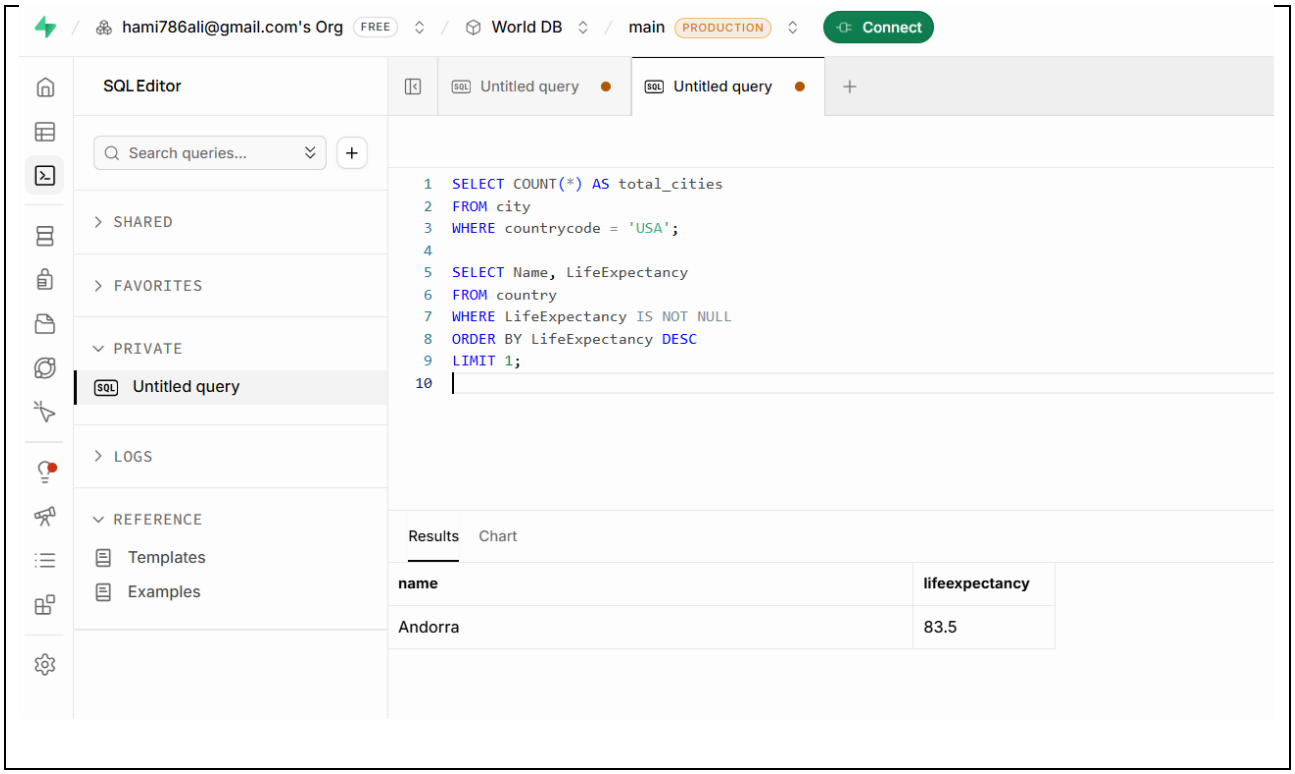
1. **Visit your Superbase profile**
2. **Create a new project**
3. **Open and copy SQL script here Create World DB on Supabase**
4. **Go to SQL Editor and paste it on new query window.**
5. **Go to Table editor and verify city, country and countrylanguage tables are created.**
6. **You can also watch this video to create it. How to create World DB on Supabase.webm**

1. **Count Cities in USA:** *Scenario:* You've been tasked with conducting a demographic analysis of cities in the United States. Your first step is to determine the total number of cities within the country to provide a baseline for further analysis.

```
SELECT COUNT(*) AS total_cities FROM city WHERE countrycode = 'USA';
```

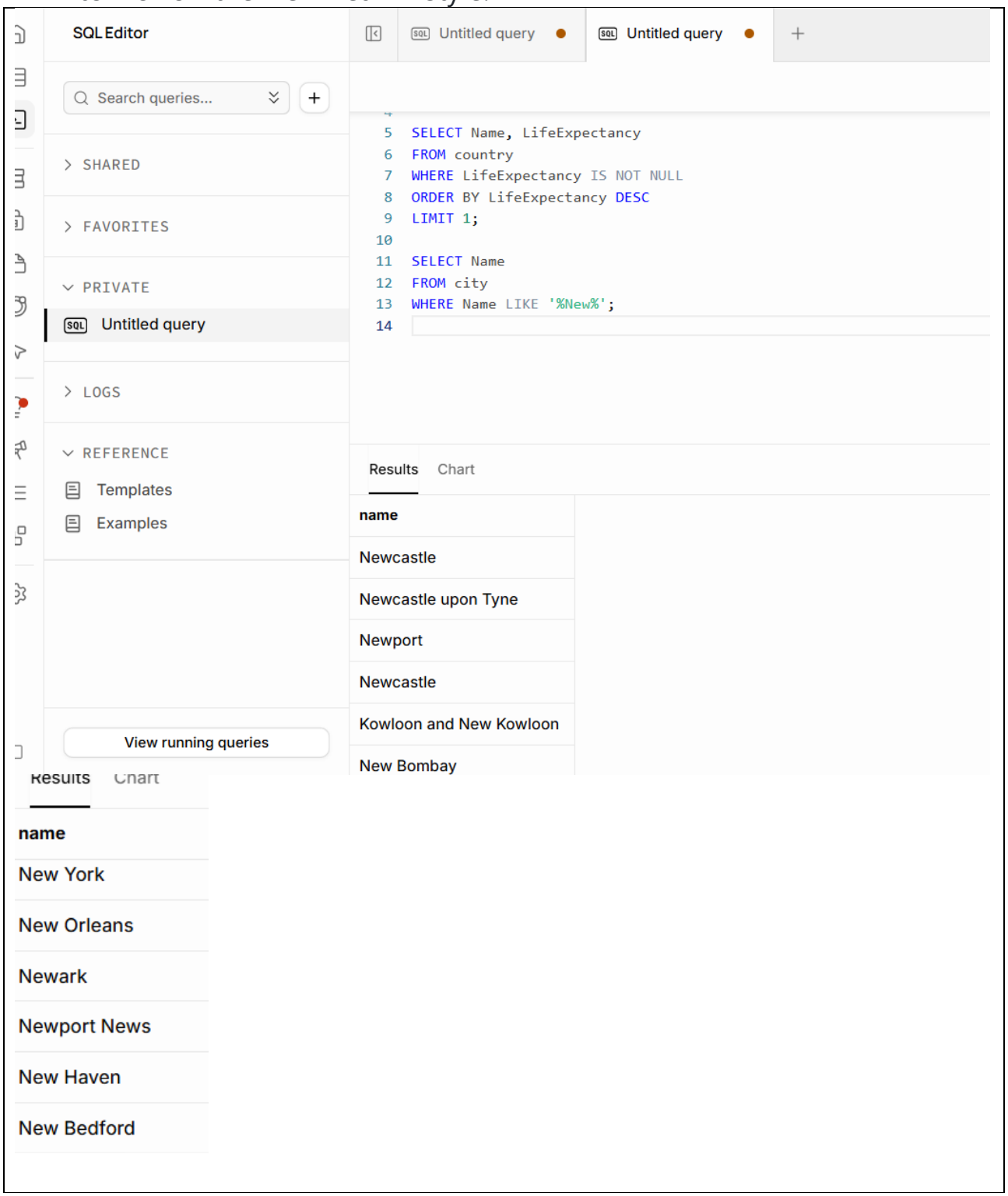


2. **Country with Highest Life Expectancy:** *Scenario:* As part of a global health initiative, you've been assigned to identify the country with the highest life expectancy. This information will be crucial for prioritising healthcare resources and interventions.

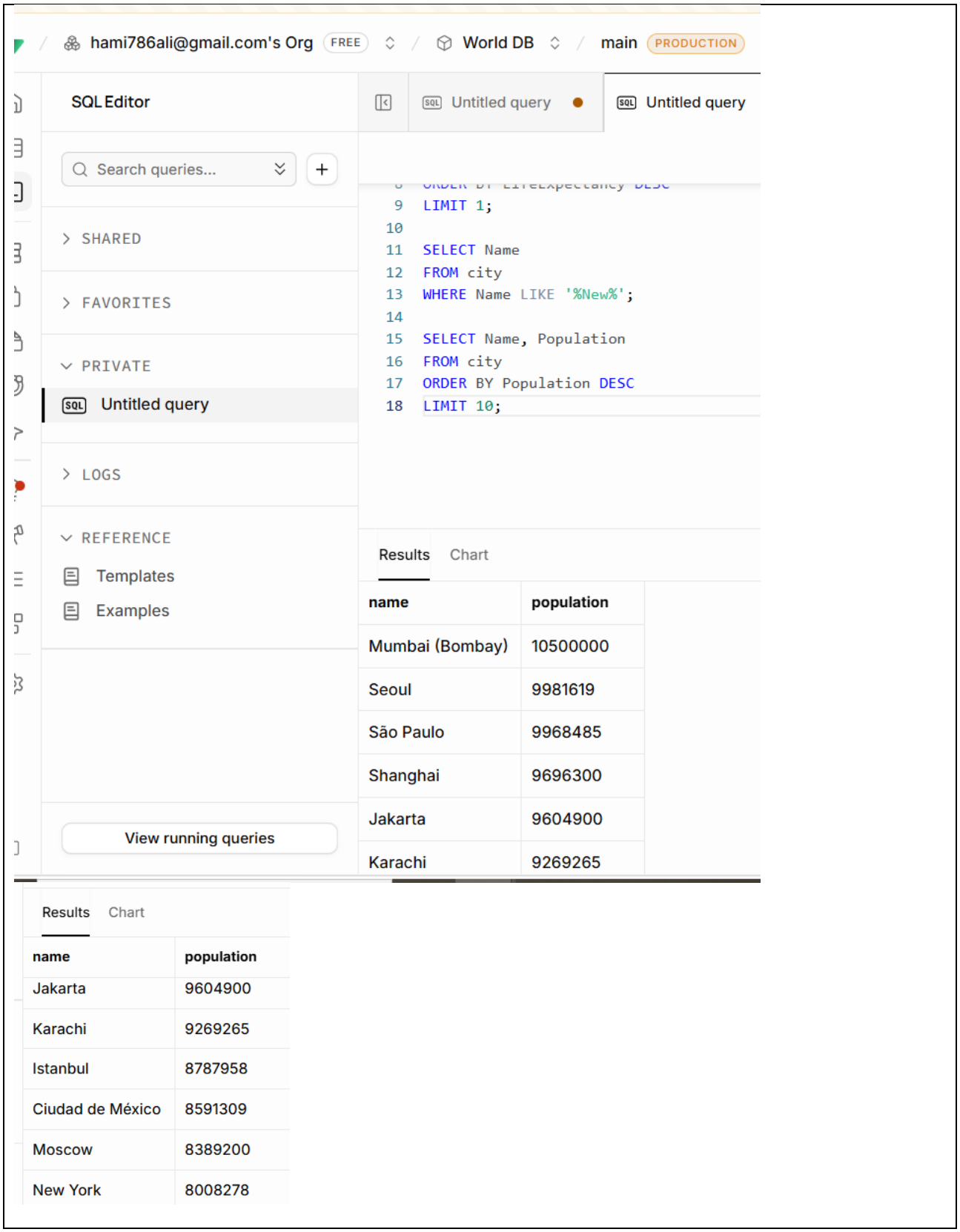


3. **"New Year Promotion: Featuring Cities with 'New :** *Scenario:* In anticipation of the upcoming New Year, your travel agency is gearing up for a special promotion featuring cities with names including the word 'New'. You're tasked with swiftly compiling a list of all cities from

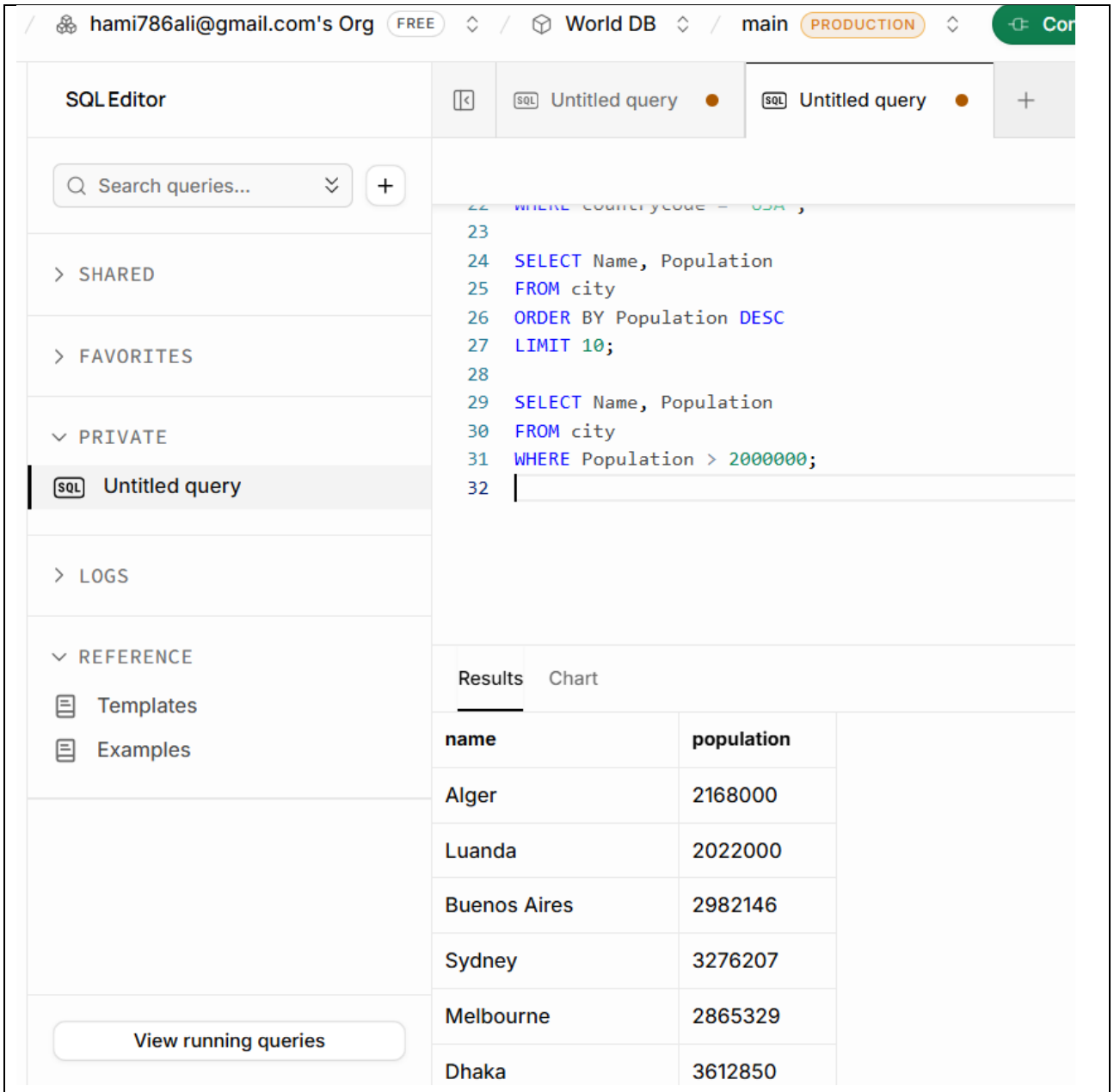
around the world. This curated selection will be essential in creating promotional materials and enticing travellers with exciting destinations to kick off the New Year in style.



4. **Display Columns with Limit (First 10 Rows):** *Scenario:* You're tasked with providing a brief overview of the most populous cities in the world. To keep the report concise, you're instructed to list only the first 10 cities by population from the database.


hami786ali@gmail.com's Org
FREE
World DB
main
PRODUCTION
SQL Editor
Untitled query
Untitled query
Search queries...
SHARED
FAVORITES
PRIVATE
Untitled query
LOGS
REFERENCE
Templates
Examples
View running queries
LIMIT 1;
SELECT Name
FROM city
WHERE Name LIKE '%New%';
SELECT Name, Population
FROM city
ORDER BY Population DESC
LIMIT 10;
Results
Chart
name
population
Mumbai (Bombay)
10500000
Seoul
9981619
São Paulo
9968485
Shanghai
9696300
Jakarta
9604900
Karachi
9269265
Results
Chart
name
population
Jakarta
9604900
Karachi
9269265
Istanbul
8787958
Ciudad de México
8591309
Moscow
8389200
New York
8008278

5. **Cities with Population Larger than 2,000,000:** *Scenario:* A real estate developer is interested in cities with substantial population sizes for potential investment opportunities. You're tasked with identifying cities from the database with populations exceeding 2 million to focus their research efforts.



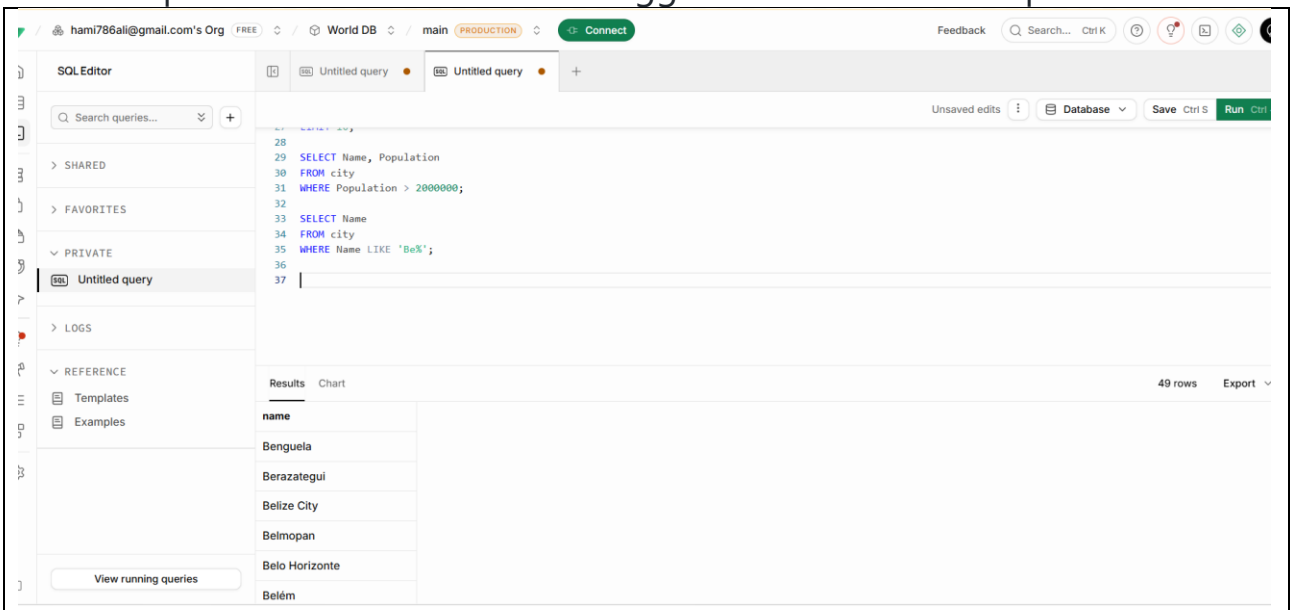
name,population
Alger,2168000
Luanda,2022000
Buenos Aires,2982146
Sydney,3276207

```
Melbourne,2865329
Dhaka,3612850
São Paulo,9968485
Rio de Janeiro,5598953
Salvador,2302832
Belo Horizonte,2139125
Fortaleza,2097757
London,7285000
Santiago de Chile,4703954
Guayaquil,2070040
Cairo,6789479
Alexandria,3328196
Giza,2221868
Madrid,2879052
Cape Town,2352121
Addis Abeba,2495000
Quezon,2173831
Jakarta,9604900
Surabaya,2663820
Bandung,2429000
Mumbai (Bombay),10500000
Delhi,7206704
Calcutta [Kolkata],4399819
Chennai (Madras),3841396
Hyderabad,2964638
Ahmedabad,2876710
Bangalore,2660088
Baghdad,4336000
Teheran,6758845
Roma,2643581
Tokyo,7980230
Jokohama [Yokohama],3339594
Osaka,2595674
Nagoya,2154376
Nairobi,2290000
Shanghai,9696300
Peking,7472000
Chongqing,6351600
Tianjin,5286800
Wuhan,4344600
Harbin,4289800
Shenyang,4265200
```

```
Kanton [Guangzhou],4256300
Chengdu,3361500
Nanking [Nanjing],2870300
Changchun,2812000
Xi´an,2761400
Dalian,2697000
Qingdao,2596000
Jinan,2278100
Hangzhou,2190500
Zhengzhou,2107200
Shijiazhuang,2041500
Santafé de Bogotá,6260862
Cali,2077386
Kinshasa,5064000
Pyongyang,2484000
Seoul,9981619
Pusan,3804522
Inchon,2559424
Taegu,2548568
La Habana,2256000
Casablanca,2940623
Ciudad de México,8591309
Rangoon (Yangon),3361700
Abidjan,2500000
Karachi,9269265
Lahore,5063499
Lima,6464693
Paris,2125246
Bucuresti,2016131
Berlin,3386667
Riyadh,3324000
Jedda,2046300
Singapore,4017733
Taipei,2641312
Bangkok,6320174
Istanbul,8787958
Ankara,3038159
Izmir,2130359
Kyiv,2624000
Toskent,2117500
Moscow,8389200
St Petersburg,4694000
```

Ho Chi Minh City,3980000
New York,8008278
Los Angeles,3694820
Chicago,2896016

6. **Cities Beginning with 'Be' Prefix:** *Scenario:* A travel blogger is planning a series of articles featuring cities with unique names. You're tasked with compiling a list of cities from the database that start with the prefix 'Be' to assist in the blogger's content creation process.



name

Benguela

Berazategui

Belize City

Belmopan

Belo Horizonte

Belém

Belford Roxo

Betim

Bento Gonçalves

Belfast

Benoni

Bekasi

Bengkulu

Belgaum

Bellary

Berhampore (Baharampur)

Beawar

Bettiah

Beerseba

Bene Beraq

Bergamo

Beppu

Beograd

Benxi

Bengbu

Bei´an

Beipiao

Beihai

Bello

Beirut

Bengasi

Beni-Mellal

Beau Bassin-Rose Hill

Benito Juárez

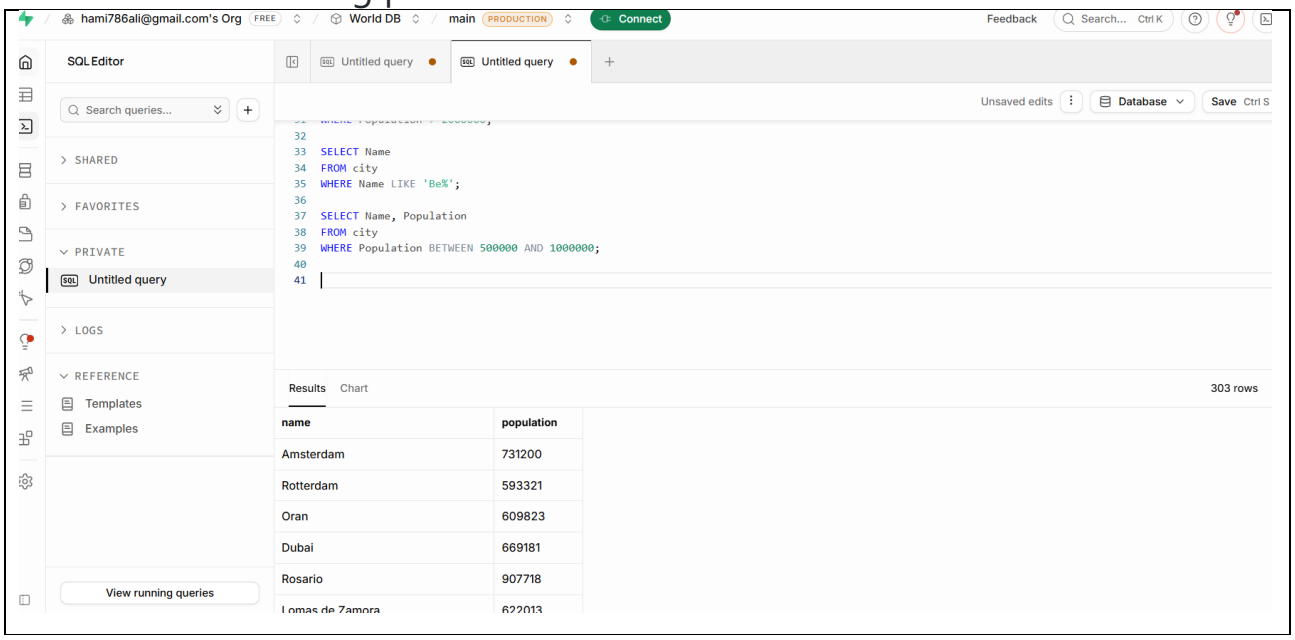
Bender (Tîghina)

Beira

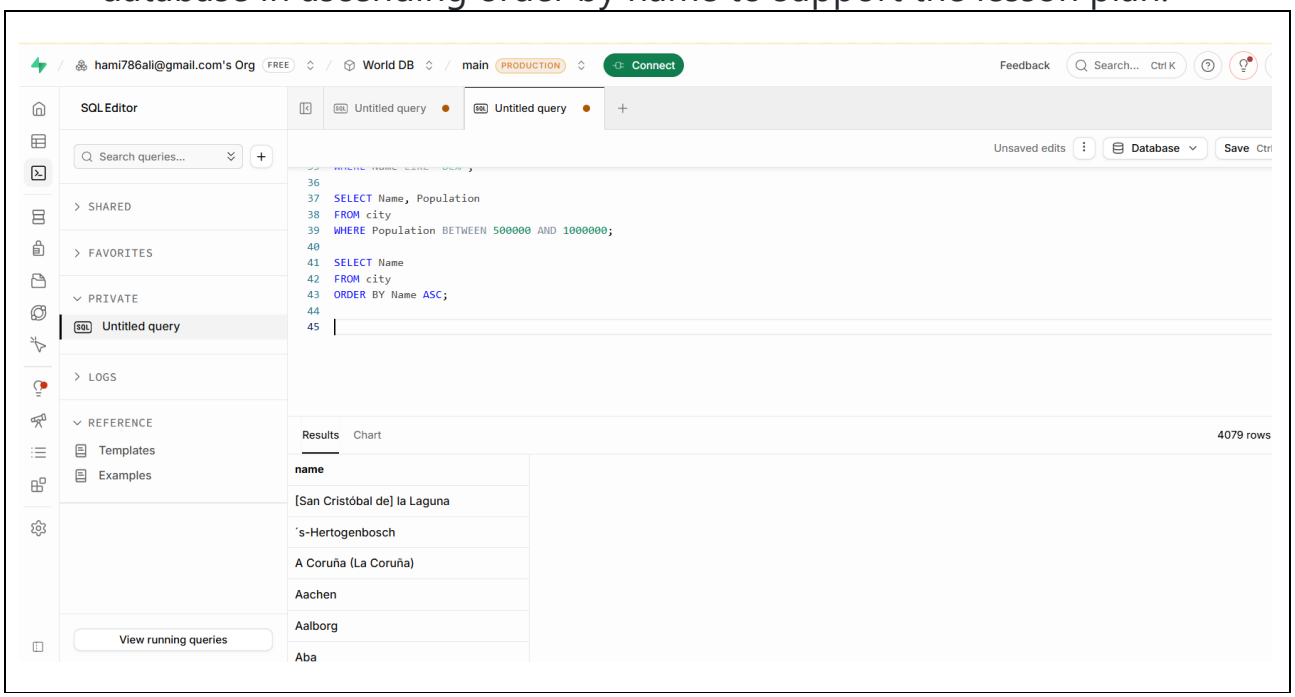
Benin City

Bergen
Besançon
Berlin
Bergisch Gladbach
Bern
Berdjansk
Berdytšiv
Belgorod
Berezniki
Beaumont
Bellevue
Berkeley

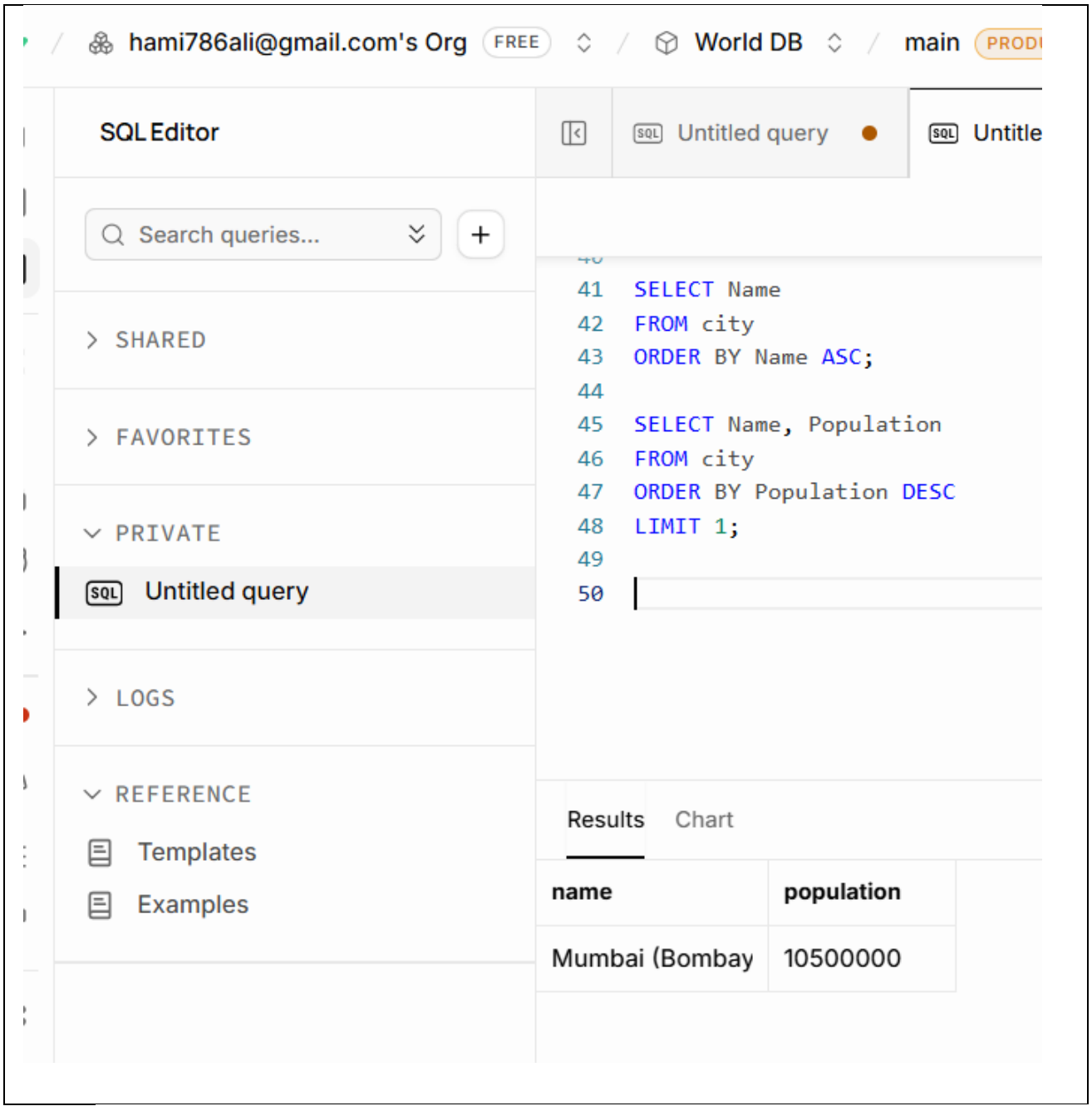
7. **Cities with Population Between 500,000-1,000,000:** *Scenario:* An urban planning committee needs to identify mid-sized cities suitable for infrastructure development projects. You're tasked with identifying cities with populations ranging between 500,000 and 1 million to inform their decision-making process.



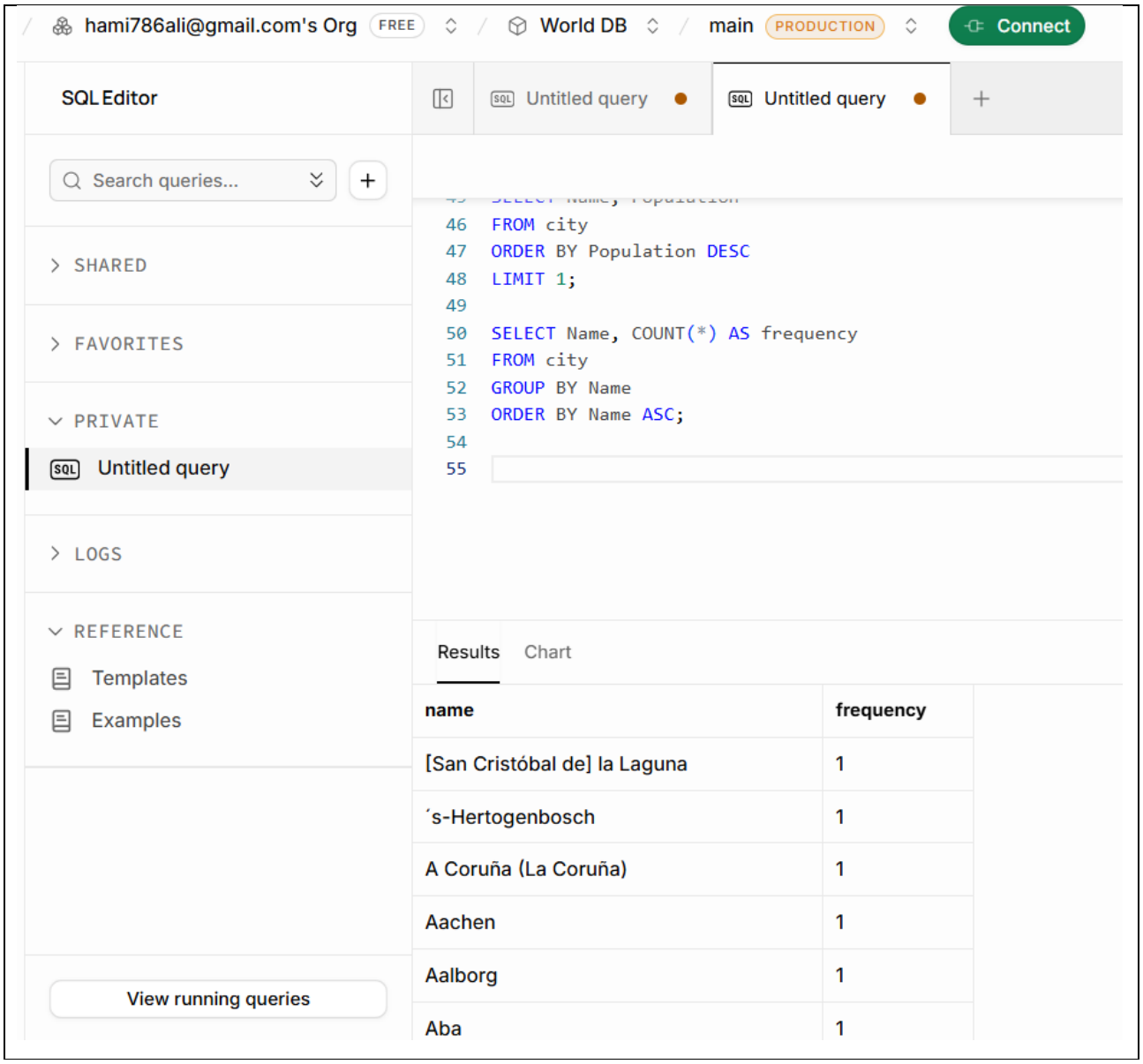
8. **Display Cities Sorted by Name in Ascending Order:** *Scenario:* A geography teacher is preparing a lesson on alphabetical order using city names. You're tasked with providing a sorted list of cities from the database in ascending order by name to support the lesson plan.



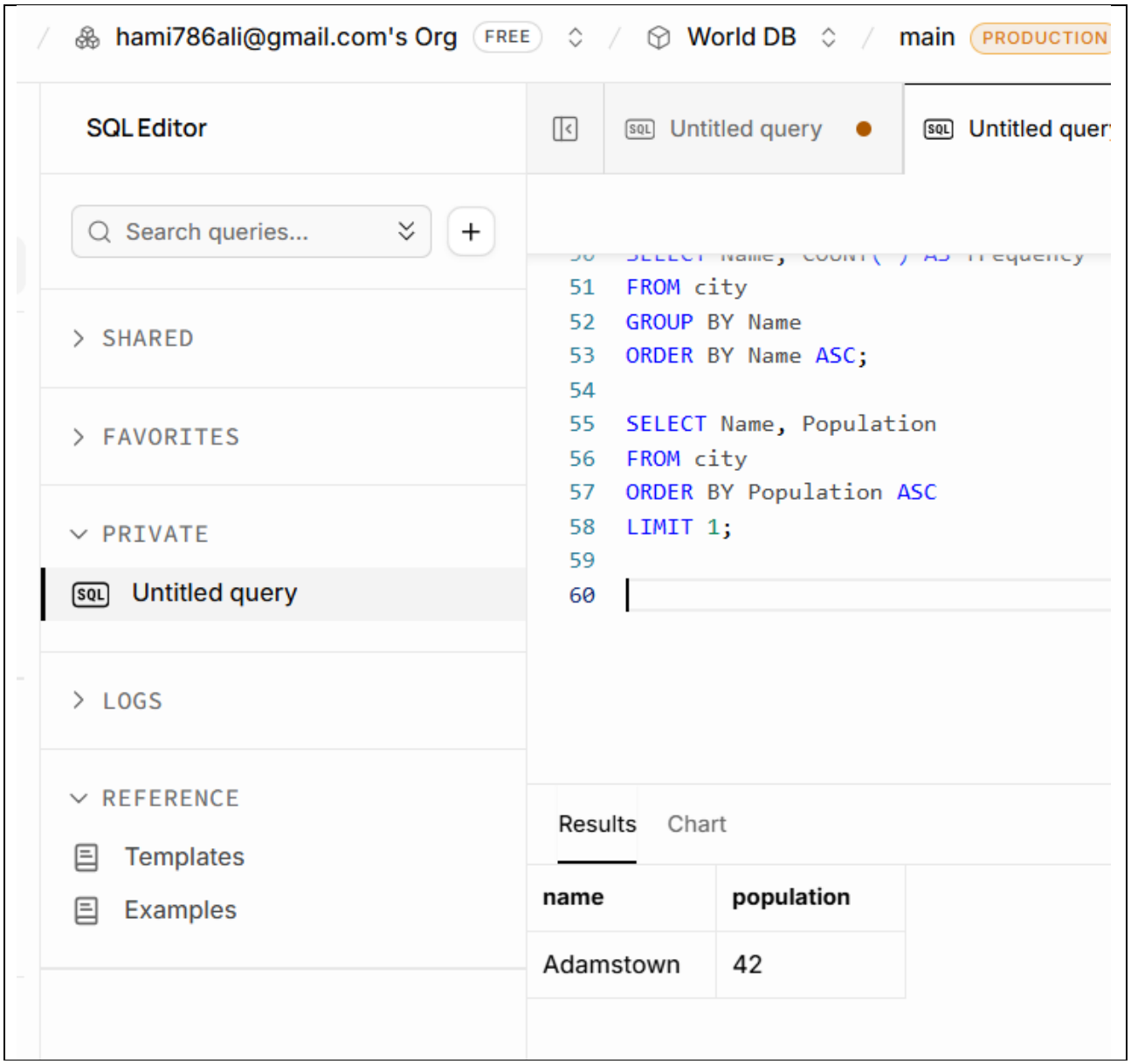
9. **Most Populated City:** *Scenario:* A real estate investment firm is interested in cities with significant population densities for potential development projects. You're tasked with identifying the most populated city from the database to guide their investment decisions and strategic planning.



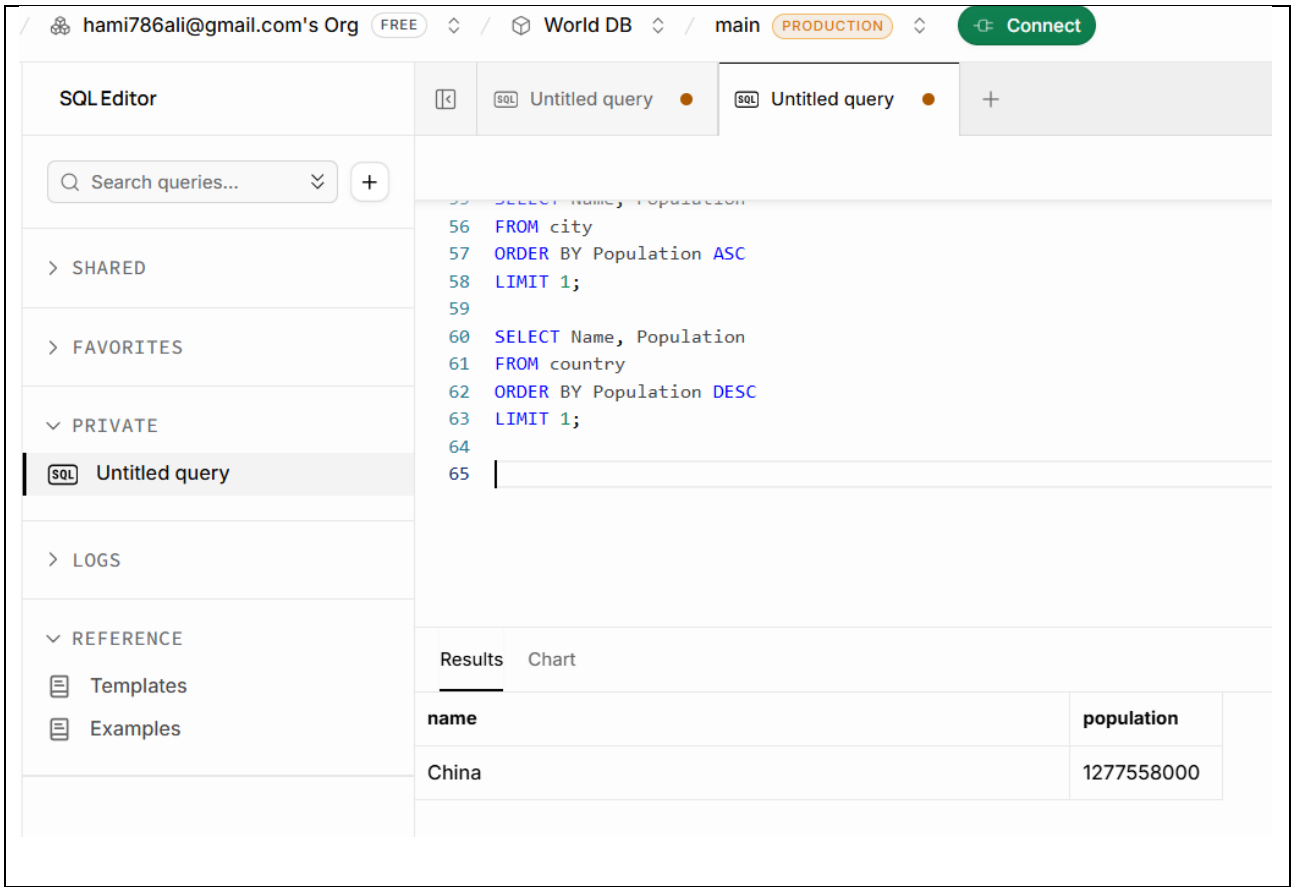
10. **City Name Frequency Analysis: Supporting Geography Education**
    Scenario: In a geography class, students are learning about the distribution of city names around the world. The teacher, in preparation for a lesson on city name frequencies, wants to provide students with a list of unique city names sorted alphabetically, along with their respective counts of occurrences in the database. You're tasked with this sorted list to support the geography teacher.



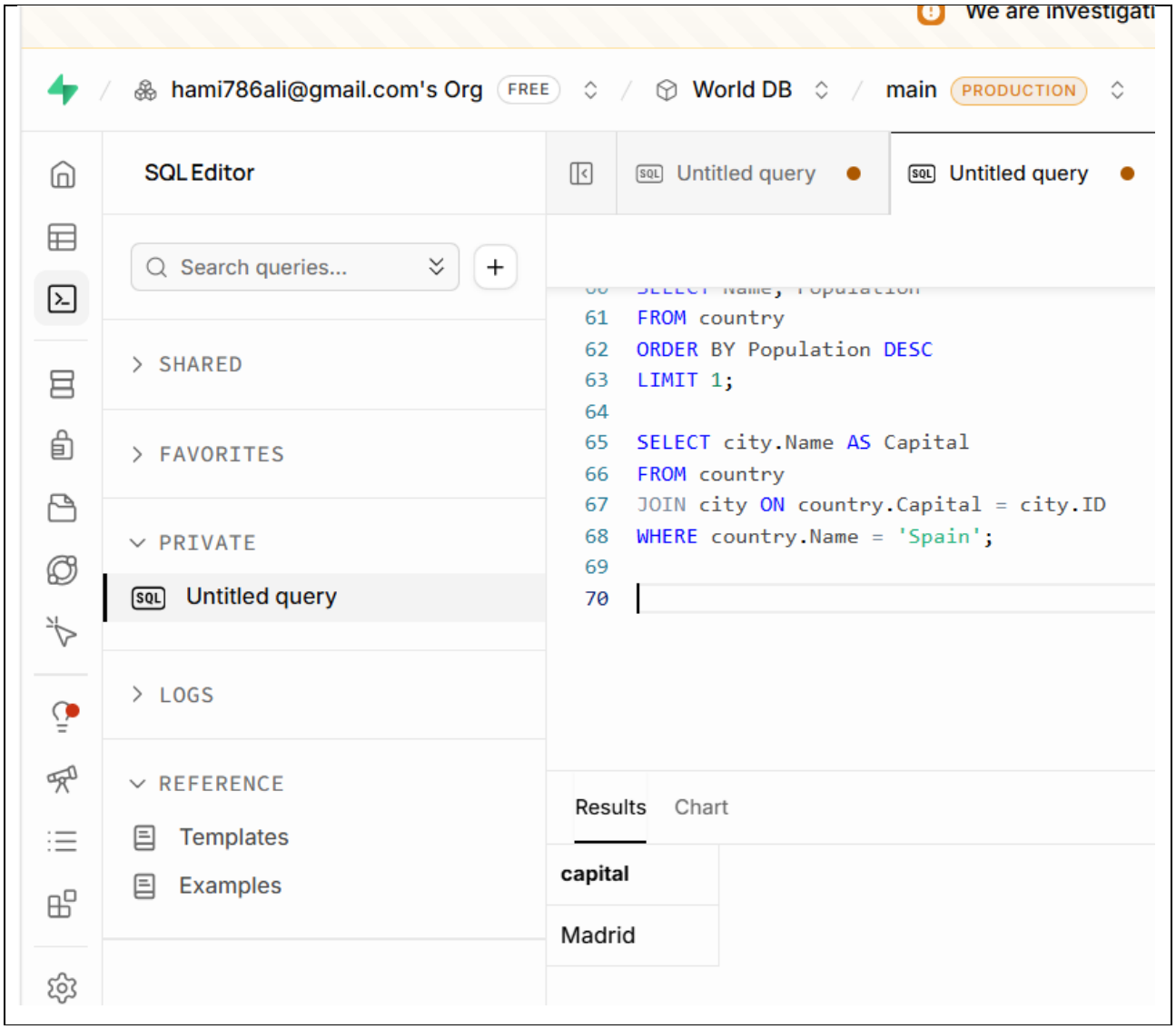
11. **City with the Lowest Population:** *Scenario:* A census bureau is conducting an analysis of urban population distribution. You're tasked with identifying the city with the lowest population from the database to provide a comprehensive overview of demographic trends.



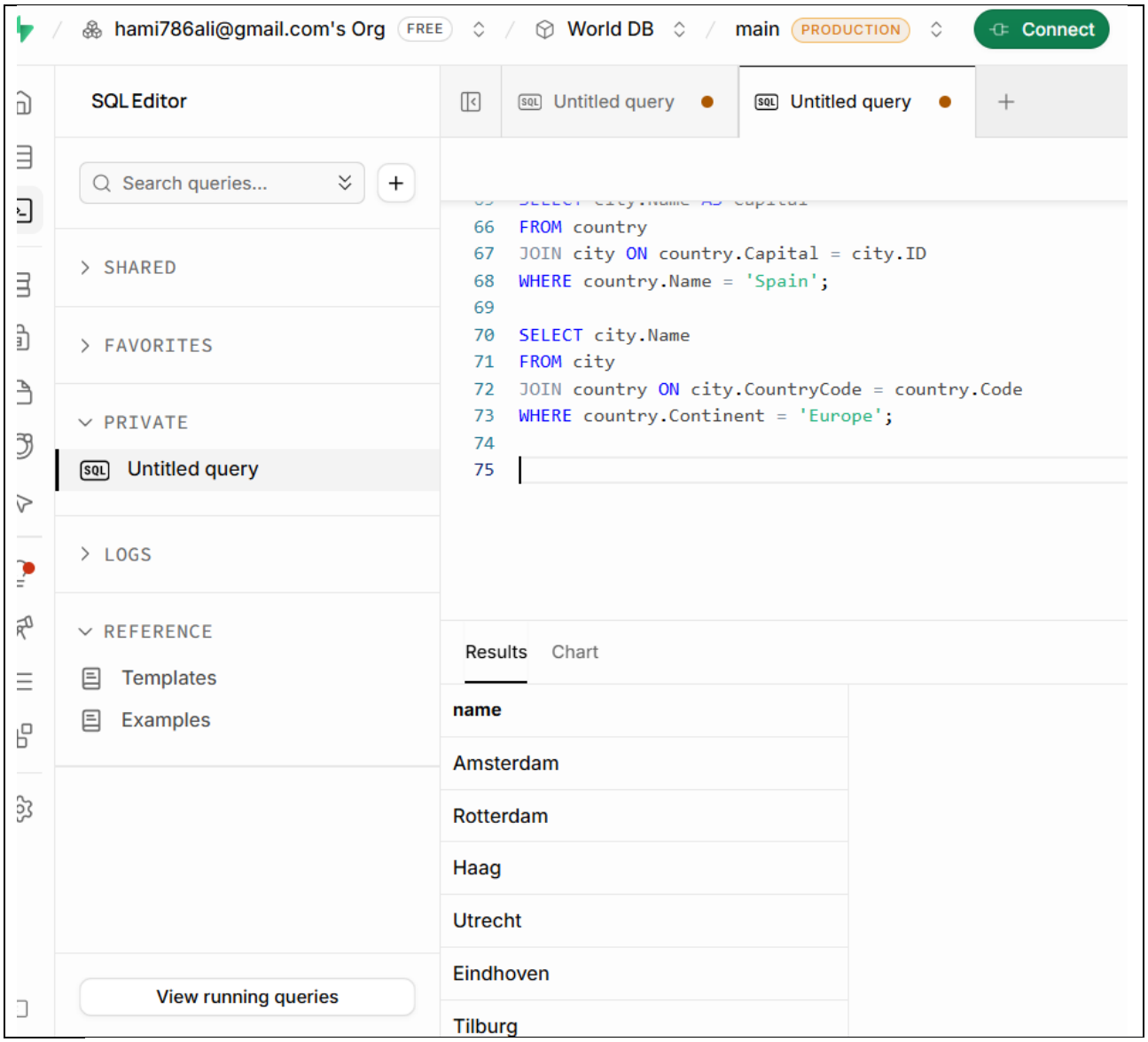
12. **Country with Largest Population:** *Scenario:* A global economic research institute requires data on countries with the largest populations for a comprehensive analysis. You're tasked with identifying the country with the highest population from the database to provide valuable insights into demographic trends.



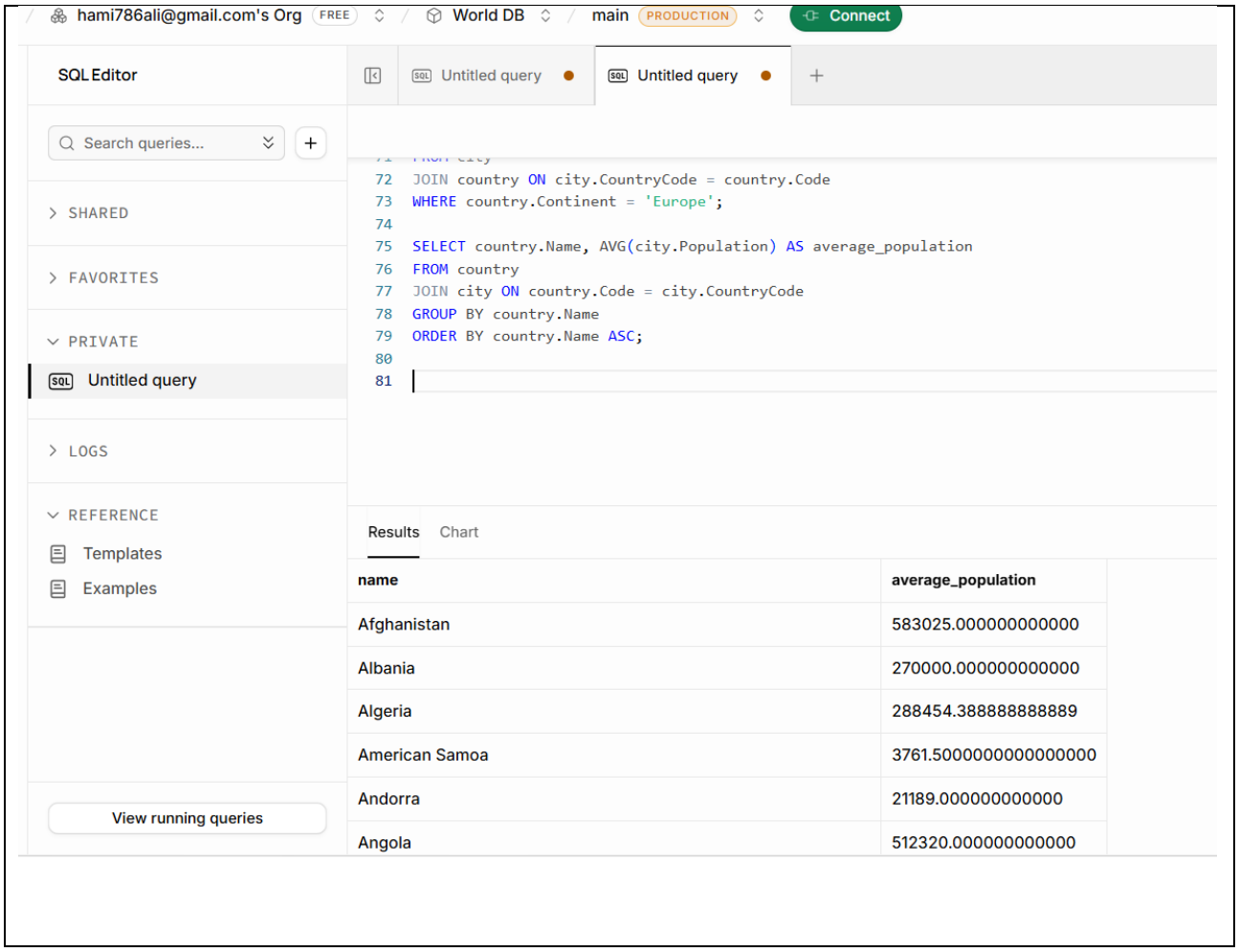
13. **Capital of Spain:** *Scenario:* A travel agency is organising tours across Europe and needs accurate information on capital cities. You're tasked with identifying the capital of Spain from the database to ensure itinerary accuracy and provide travellers with essential destination information.



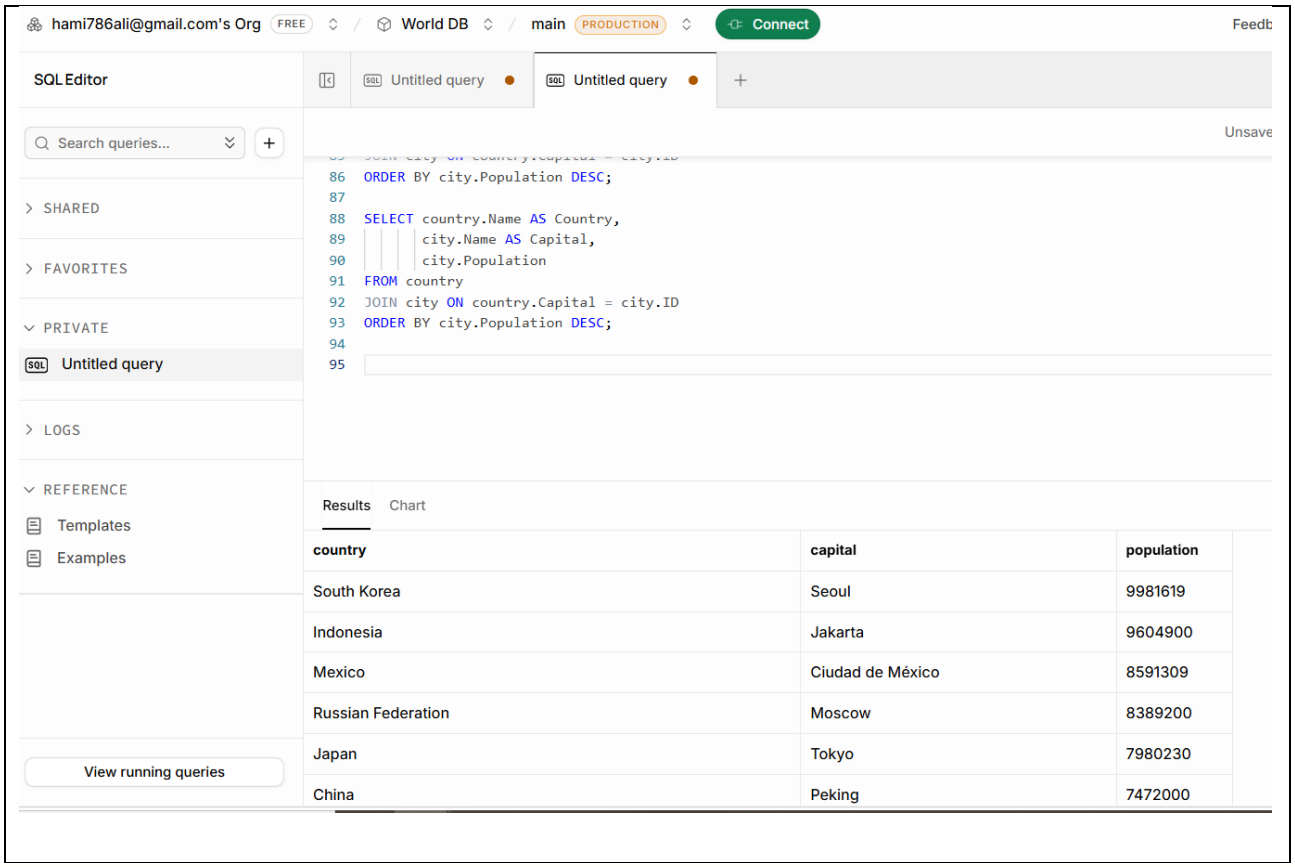
14. **Cities in Europe:** *Scenario:* A European cultural exchange program is seeking to connect students with cities across the continent. You're tasked with compiling a list of cities located in Europe from the database to facilitate program planning and student engagement.



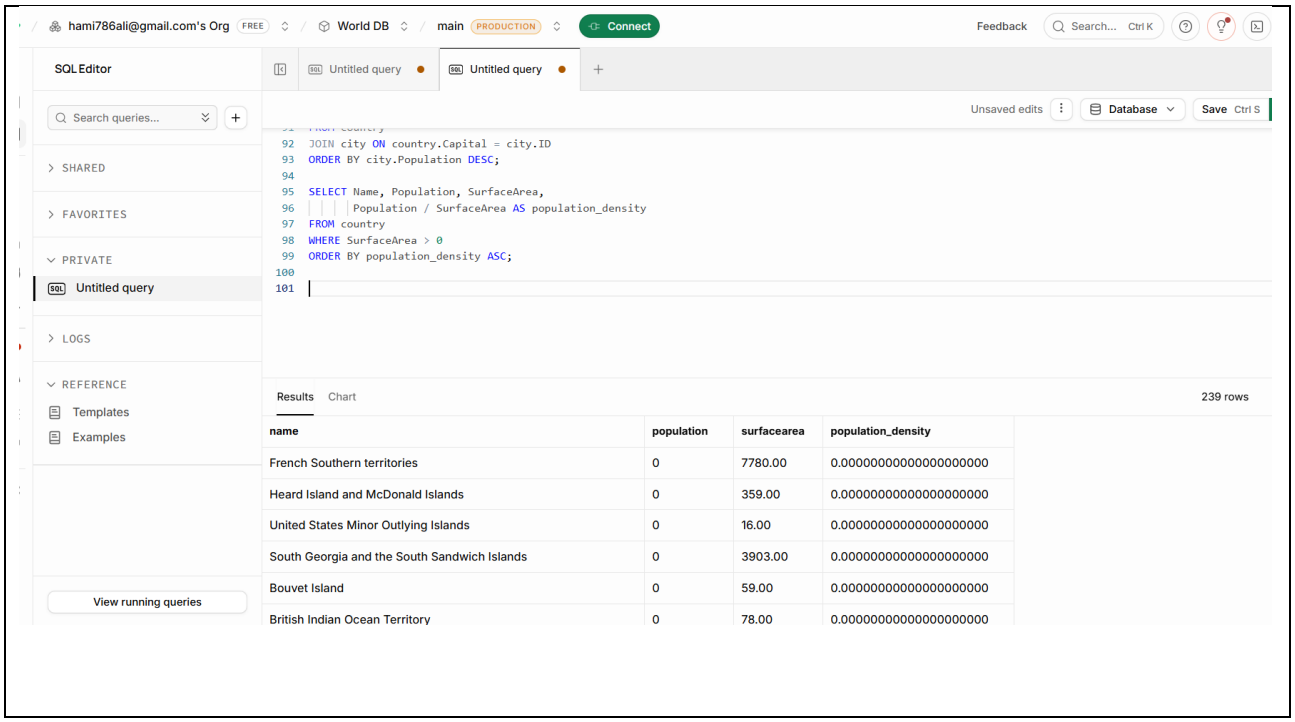
15. **Average Population by Country:** *Scenario:* A demographic research team is conducting a comparative analysis of population distributions across countries. You're tasked with calculating the average population for each country from the database to provide valuable insights into global population trends.



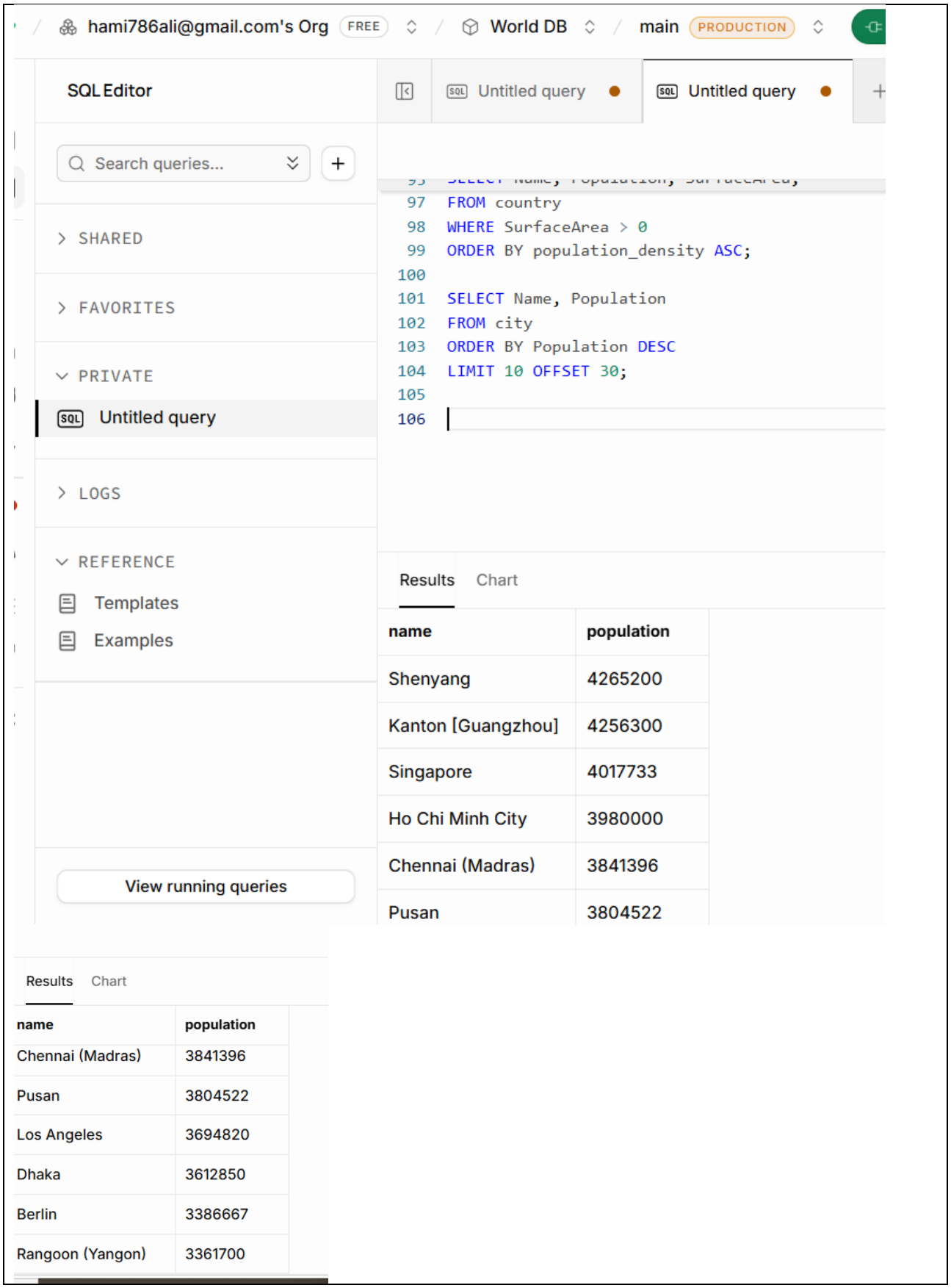
16. **Capital Cities Population Comparison:** *Scenario:* A statistical analysis firm is examining population distributions between capital cities worldwide. You're tasked with comparing the populations of capital cities from different countries to identify trends and patterns in urban demographics.



17. **Countries with Low Population Density:** *Scenario:* An agricultural research institute is studying countries with low population densities for potential agricultural development projects. You're tasked with identifying countries with sparse populations from the database to support the institute's research efforts.



18. **Display Columns with Limit (Rows 31-40):** *Scenario:* A market research firm requires detailed information on cities beyond the top rankings for a comprehensive analysis. You're tasked with providing data on cities ranked between 31st and 40th by population to ensure a thorough understanding of urban demographics.


hami786ali@gmail.com's Org
FREE
World DB
main
PRODUCTION
SQL Editor
Untitled query
Untitled query
Search queries...
SHARED
FAVORITES
PRIVATE
Untitled query
LOGS
REFERENCE
Templates
Examples
View running queries
97 FROM country
98 WHERE SurfaceArea > 0
99 ORDER BY population_density ASC;
100
101 SELECT Name, Population
102 FROM city
103 ORDER BY Population DESC
104 LIMIT 10 OFFSET 30;
105
106
Results
Chart
name
population
Shenyang
4265200
Kanton [Guangzhou]
4256300
Singapore
4017733
Ho Chi Minh City
3980000
Chennai (Madras)
3841396
Pusan
3804522
Results
Chart
name
population
Chennai (Madras)
3841396
Pusan
3804522
Los Angeles
3694820
Dhaka
3612850
Berlin
3386667
Rangoon (Yangon)
3361700

Additional Resources

1) DataCamp Recommended Courses
    A. Introduction to SQL - DataCamp Learn
    B. Intermediate SQL - DataCamp Learn
    C. Joining Data in SQL - DataCamp Learn

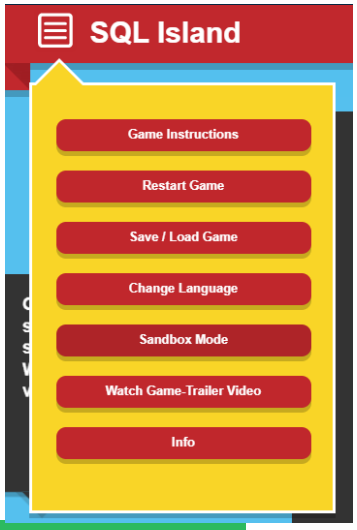
2) SQLBolt - Learn SQL - Introduction to SQL : Welcome to SQLBolt, a series of interactive lessons and exercises designed to help you quickly learn SQL right in your browser.

3) MySQL Tutorial: Tutorial website.

4) MySQL Basics Tutorial Series - YouTube: MySQL Basics Tutorial Series

5) Hackerrank Challenges: How to use hackerrank challanges.odt

6) SQL Island: After the survived plane crash, you will be stuck on SQL Island for the time being. By making progress in the game, you will find a way to escape from this island.
Change Language to English.



Course Notes

It is recommended to take notes from the course, use the space below to do so, or use the revision guide shared with the class:

We have included a range of additional links to further resources and information that you may find useful, these can be found within your revision guide.

END OF WORKBOOK

Please check through your work thoroughly before submitting and update the table of contents if required.

Please hand in your workbook in MS Assignment page.